

Peripheral Nervous System “Neural Conduction, and Neurotransmitters”

Integrated Biomedical and Clinical Sciences OMFS 512

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Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Objectives

1. Describe steps involved in sensory transduction and action potential generation
2. Describe the autonomic nervous system and understand the role of different neurotransmitters.
3. Compare and contrast different types of seizures as well as management strategies
4. Describe the clinical manifestations and management of Parkinson's disease

Outline

- Somatosensory function: Pain physiology
- Autonomic nervous system.
- Different types of neurotransmitters.
- Pathological conditions due to neurotransmitter deficiency.

Stages of Pain Physiology

- **Transduction**

Painful stimuli will generate action potential at the nerve ending

- **Transmission**

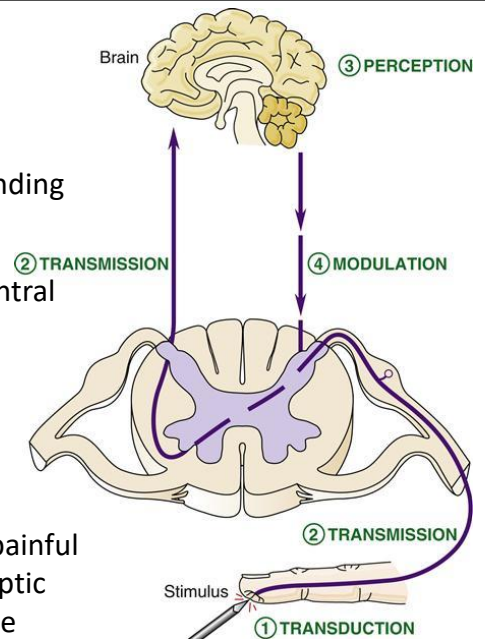
Transfer of the sensory signal from the peripheral to the central nervous system

- **Perception**

Pain recognition process by the brain

- **Modulation**

The brain and spinal cord aim to decrease or suppress the painful sensation either by motor modulation, decreasing the synaptic transmission, or secreting endogenous endorphins from the pituitary gland.



Activation of Nociceptors

Painful stimulus/tissue injury 



Activates the release of mediators like bradykinin, substance p, histamine, etc.



Activates the ion gated transient receptor potential (TRP) channels on the axon terminals of 1st order sensory neurons (pseudounipolar neurons).



Increases membrane permeability to sodium and calcium ions.



Excitation of the afferent nerve ending (action potential) of the spinothalamic pathway.

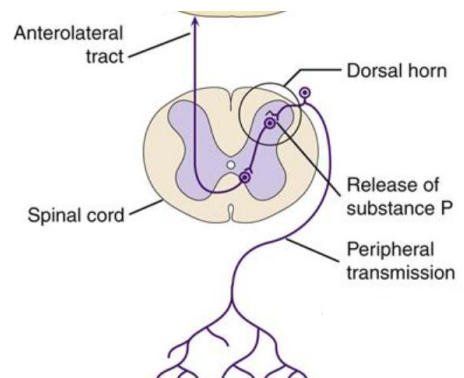
Reference: Transient receptor potential ion channels in primary sensory neurons as targets for novel analgesics. *Valente et al.*, 2014.

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Conduction at the Spinal Cord

- In the dorsal horn of the spinal cord, the A δ fibers release glutamate and C-fibers release glutamate and substance P.
- This transfers the pain signal to the second order neurons and afferent fibers then , thalamus and the somatosensory cortex via modulating the membrane potential of neurons.



How are signals transmitted in the Nervous system?

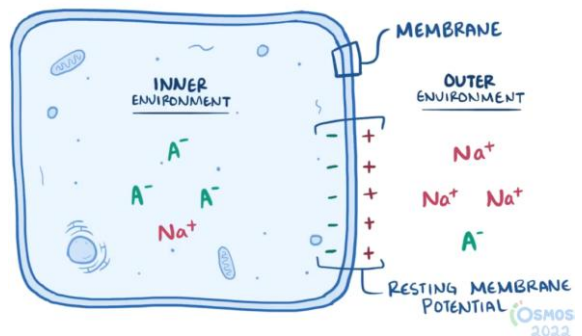
Neural Conduction

Neurons respond to electrical, chemical, or mechanical stimuli by producing changes in ion conduction across the cell membrane

Membrane potential

- A membrane potential is caused by the difference of ion concentration across the cell membrane.
- The resting membrane potential is an equilibrium state that results in a significant difference in voltage across the neural membrane.
- The resting membrane potential for neurons is **-70 mV**.

- At rest, the cell membrane is freely permeable to potassium ions but relatively impermeable to sodium ions and other intracellular anions (A^-).
- In neurons, the concentration of K^+ is higher inside than outside the cell, while the reverse is the case for Na^+ . This concentration difference is established by Na, K ATPase pump.



Action Potential



Rapid (7 millisecond), all-or-none signal that causes a change in the membrane polarization, followed by a return to the resting membrane potential.

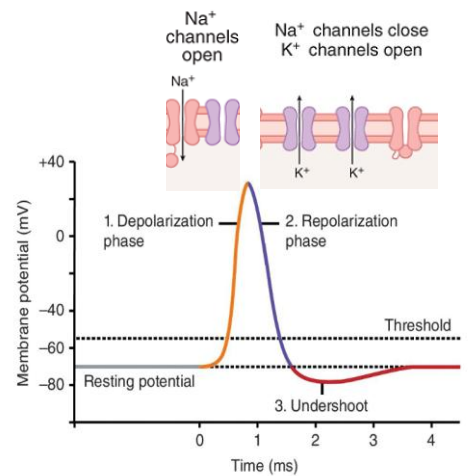
It causes the activation of voltage-gated sodium channels and starts a chain reaction that propagates the electrical charge along the axon.

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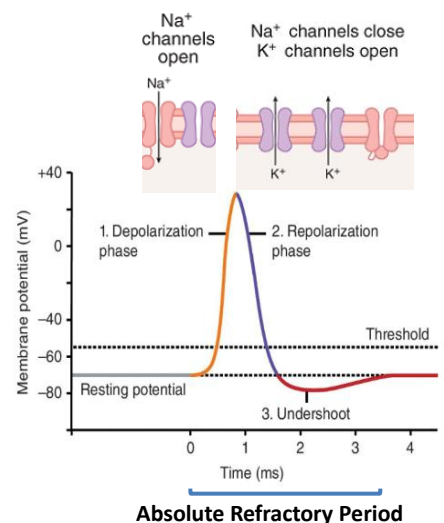
Resting Stage= resting membrane potential before the action potential begins (freely permeable to potassium ions). The membrane is “polarized” (-70 mv).

Depolarization Stage= upon excitation, sod. channel are open, influx of sodium ions increase the membrane potential to the positive direction.



Repolarization Stage= After reaching a threshold of +40 mV, sodium channels close, and potassium channels are actively open. Rapid outflux of potassium ions establishes the membrane repolarization.

Refractory Period= starting from depolarization till reaching the resting membrane potential, the ion channels will not respond to any new stimuli.

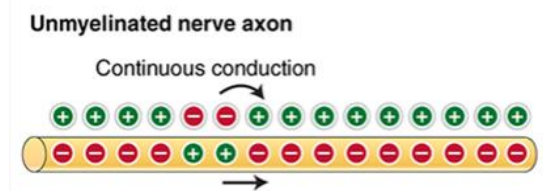


Conduction Velocity

Depends on:

- Fiber diameter: larger diameter of the nerve fiber= lower internal resistance = faster conduction.
- Myelination: myelin acts as an insulator and increases conduction velocity.

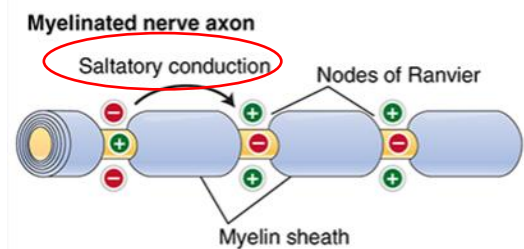
Non-myelinated axon: the action potential is conducted via continuous conduction through voltage gated ion channels. Demyelination diseases significantly impede nerve conduction.



Myelinated axons: conduction speed is increased by myeline. Action potential move via discrete jumps from node-to-node. The voltage-gated Na^+ channels are only expressed at the nodes of Ranvier (saltatory conduction).

Saltatory conduction adv:

- 1- Saves energy by decreasing the use of sodium/potassium pump in the axon
- 2- Increase speed of nerve impulse leading to fast neural communication.



Nodes of Ranvier: The region of the axon between consecutive myelin sheaths.

A single neuron can receive stimuli from multiple neurons at the same time.

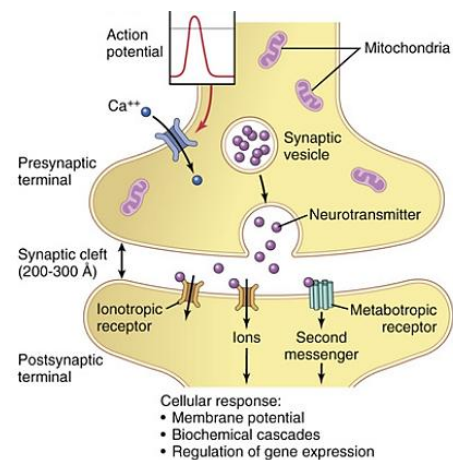
- **Excitatory postsynaptic potential (EPSP):** stimulation that causes membrane depolarization of the postsynaptic neurons (opening the sodium channels).
- **Inhibitory postsynaptic potential (IPSP):** stimulus that causes a membrane hyperpolarization, i.e inhibit an action potential by activating the chloride channels and causing a more negative membrane potential.

Propagation of Action Potential between neurons

synaptic Transmission

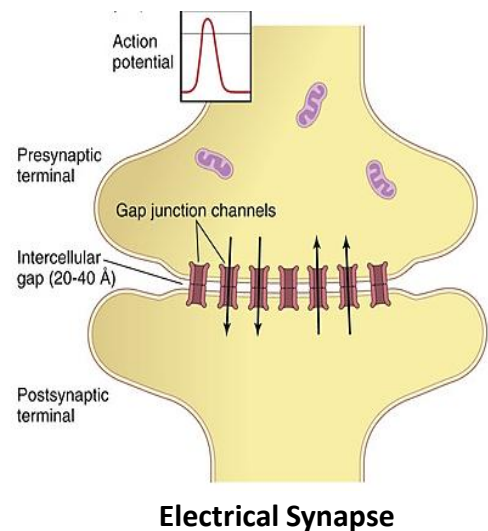
Synaptic transmission: axon terminal of one neuron propagate the signal to the dendrites of another neuron.

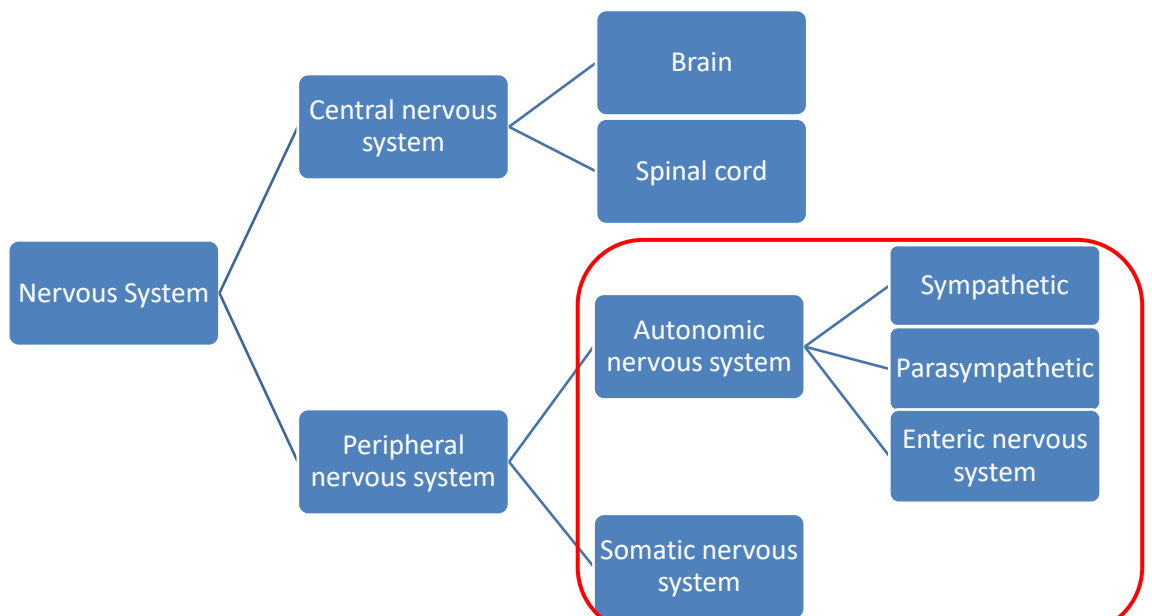
1- Chemical synapses “Common” neurotransmitter include acetylcholine, norepinephrine, epinephrine, histamine, gamma-aminobutyric acid (GABA), glycine, serotonin, and glutamate.



Chemical Synapse via neurotransmitters

2- Electrical synapse: transfer of action potential across gap junctions. The gap junctions allow free movement of ions from one cell to the next. Allow extremely rapid neural communication. Play an essential role in the development and function of the nervous system.





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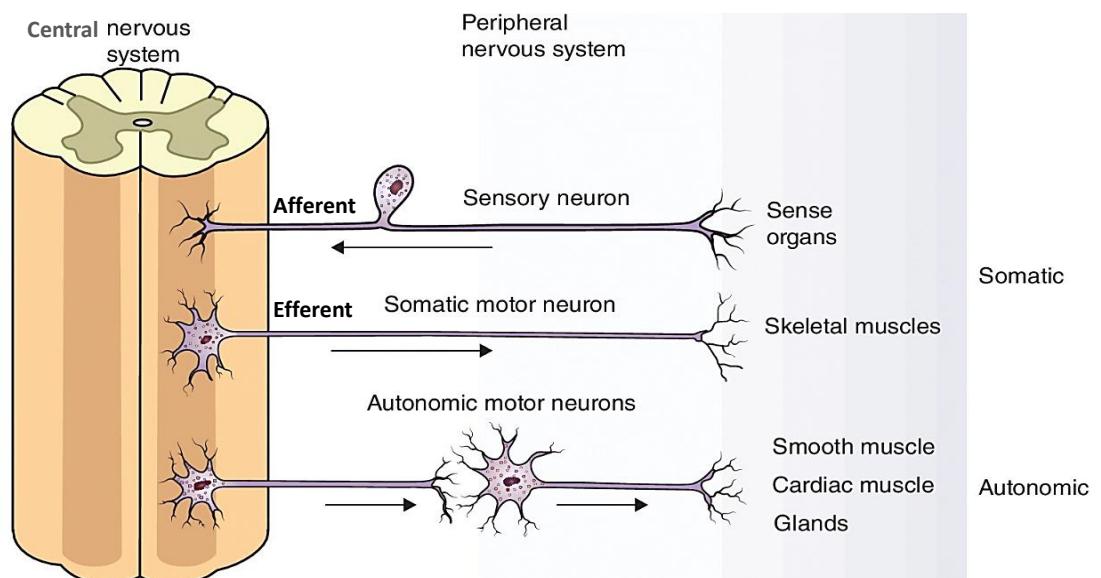
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Functionally, PNS is subdivided into:

- **Afferent (sensory) nerves:** Sensory nerves synapse at the dorsal root of the spinal cord. Pseudounipolar.
- **Efferent (motor) nerves:** Motor nerves project to effector organ through the ventral horn. Acetylcholine is released at the neuromuscular junction. Neurons are multipolar.

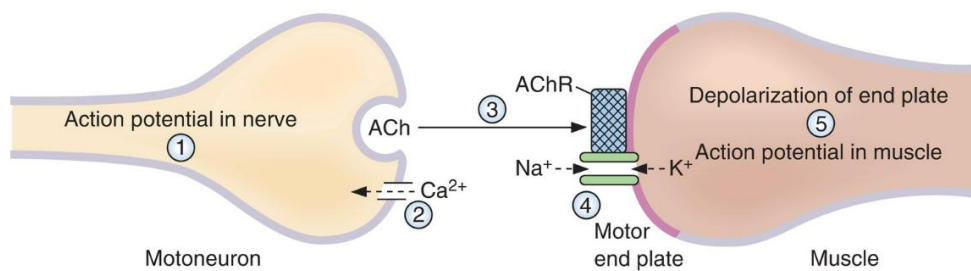
Somatic nervous system, responsible for voluntary control. Provides sensory and motor innervation to the body, except the viscera in the body cavities, smooth muscle, and glands.

Autonomic nervous system, responsible for internal functions of the body functions, or involuntary functions (visceral efferent and afferent fibers). Maintain **homeostasis**. Consist of a relay of 2 neurons.



Somatic Nervous System

- Responsible for voluntary control.
- At the skeletal muscle, axons of motor neurons release acetylcholine at the neuromuscular junction leading to muscle action.



Autonomic Nervous System

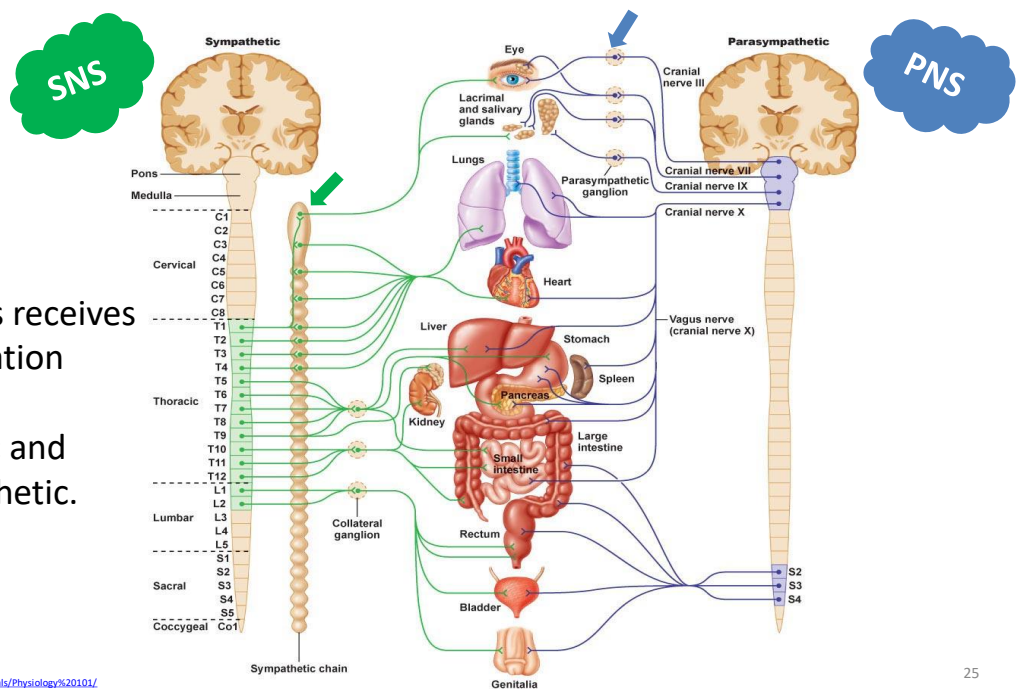
Functions:

Main purpose is to maintain homeostasis and integrate the stress response.

- It controls involuntary functions.
- Regulates of blood pressure
- GI responses to food.
- Bladder contraction of the urinary bladder.

- **Enteric nervous system:** an extensive web-like structure that can function independently of the remainder of the nervous system. It contains over 100 million neurons, and it is responsible for the regulation of digestive processes.
- **Sympathetic nervous system:** Fight or Flight reactions and exercise.
- **Parasympathetic nervous system:** Rest and Digest reactions.

- Most organs receive dual innervation from the sympathetic and parasympathetic.



Adapted from:
http://drouais.faculty.mjc.edu/Course%20Materials/Physiology%20101/Chapter%20Notes/Fall%202007/chapter_11%20Fall%202007.html

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Central Control of ANS

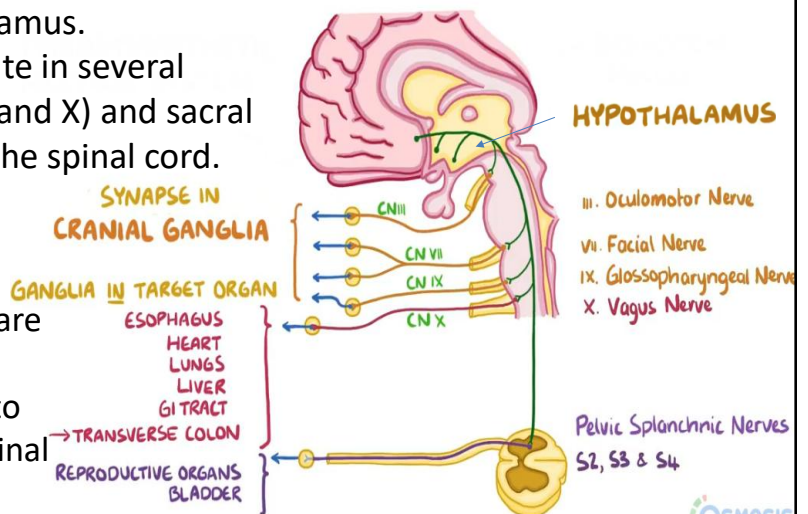
Influence over autonomic output is maintained by autonomic centers in several locations of the brain. For example:

- Hypothalamus: Largest center. Receives information from different receptors in the body. Hypothalamus integrates information and send output down the spinal cord, via hypothalamospinal tract, to engage the preganglionic fibers of SNS and PNS.
- Vasomotor and vasodilator centers in the medulla.
- Respiratory centers are in the medulla and pons.

Parasympathetic nervous system

Controlled from the hypothalamus.
Preganglionic neurons originate in several cranial regions (CN III, VII, IX, and X) and sacral segments (S2, S3, and S4) of the spinal cord.

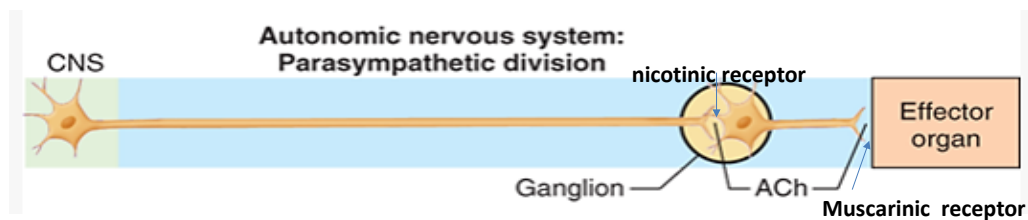
Around 75% of all parasympathetic nerve fibers are in the Vagus nerve (CN X) that send parasympathetic output to the entire thoracic and abdominal regions.



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- Cholinergic preganglionic efferent fibers release acetylcholine to activate the nicotinic receptors (ion gated channels) of the postganglionic fibers.
- Post-ganglionic fibers release acetylcholine to activate muscarinic (G protein coupled) receptors of the effector organs.
- The preganglionic fibers are longer than the postganglionic fibers
- Parasympathetic promotes the normal physiological functions in several organ systems including stimulating salivary, gastric and bile release

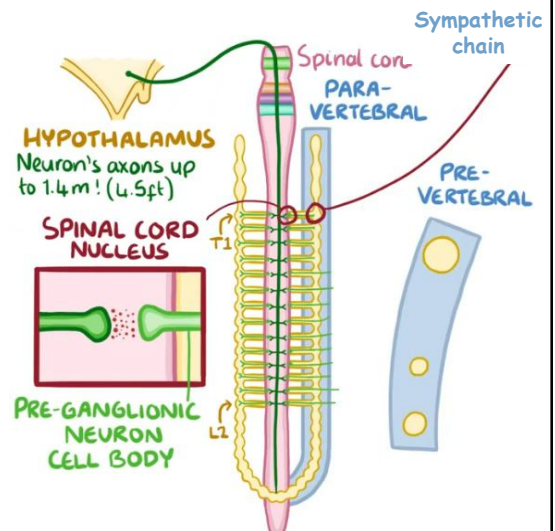


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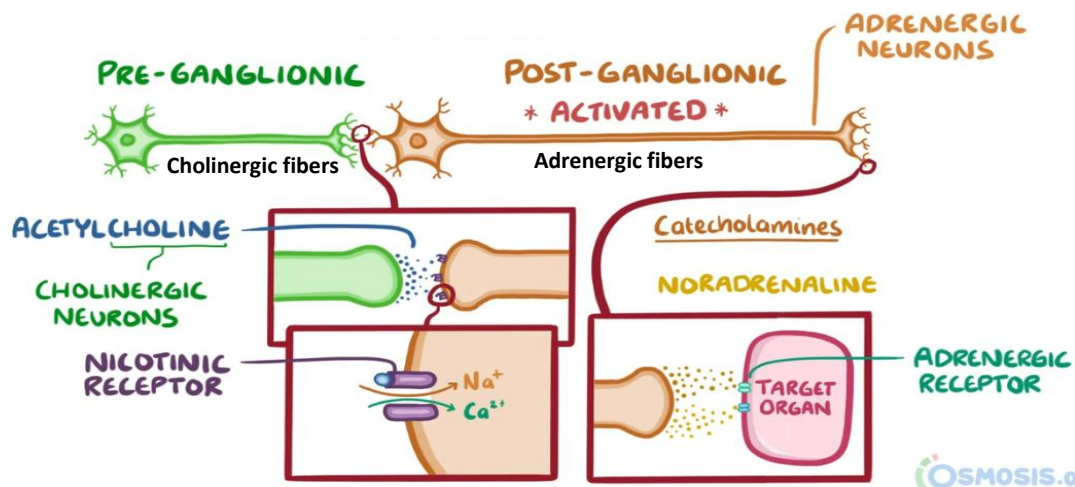
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Sympathetic nervous system

- Dominant system in a stress state or the “fight-or-flight” response.
- Hypothalamus control. Preganglionic neurons of the sympathetic system arise from the thoracic and lumbar regions of the spinal cord (from T1 to L2).
- Some preganglionic axons synapse with postganglionic neurons in sympathetic chain (paravertebral).



Sympathetic Synapses



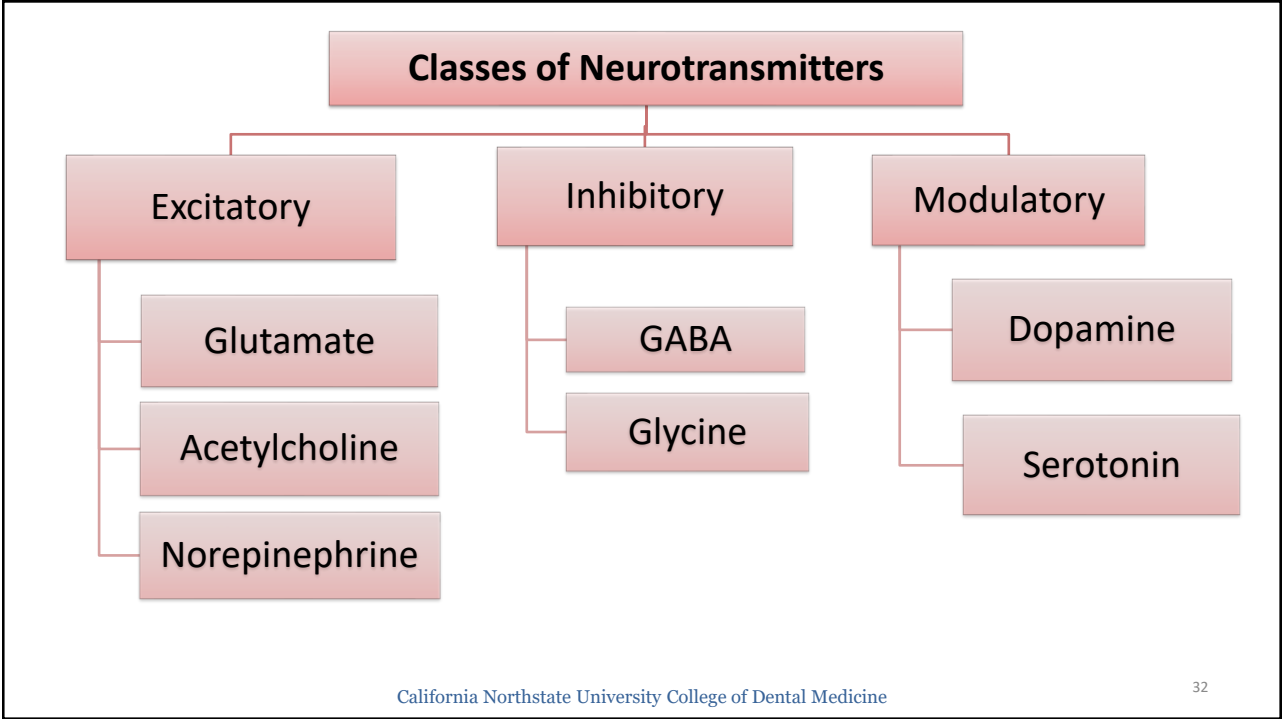
Preganglionic fibers are short and cholinergic. Postganglionic sympathetic fibers are long and adrenergic fibers.

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Neurotransmitters

Chemical compounds that transmits signals between neurons or carry the signal from the neurons to cells in the effector organs. They can be excitatory, inhibitory or both.



According to chemical structure & onset of action:

A. Small molecules, Rapid acting

1- **Amino acids** like glutamate, gamma-aminobutyric acid (GABA) and glycine. Synthesized via transamination reactions.

2- **Monoamines** like dopamine, norepinephrine, epinephrine, serotonin, and histamine.

3- **Gaseous** like nitric oxide. Diffuse freely across membrane to modulate synapsis & cause vasodilation

4- **Purines** like ATP. Can act as a transmitter or co-transmitter at synapses in PNS & CNS. Also, can modify the action of other neurotransmitters.

5- **Other** like acetylcholine.

B. Neuropeptides, slow acting: several hormones (hypothalamic releasing hormones, growth hormones, somatostatin,...) as well as substance P, endorphins. Synthesized from large polypeptides in the rough ER and transported in secretory granules to nerve terminals.

Acetylcholine



Neurotransmitter of sympathetic preganglionic neurons and the entire parasympathetic nervous system. It also affects the CNS.



Ach is synthesized by choline acetyltransferase (from Acetyl-CoA + Choline). Ach is degraded by acetylcholinesterase.



Acetylcholine mediates the effect of the parasympathetic nervous system with both muscarinic and nicotinic activity.



Pharmaceutical compounds that modulate the muscarinic and/or nicotinic receptors can mimic the effect of acetylcholine in the autonomic nervous system.

Cholinergic nerves fibers

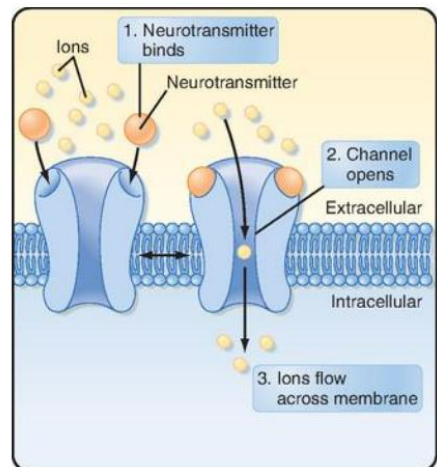
Nerve fibers that synthesize, and release acetylcholine are termed cholinergic fibers. Acetylcholine is the neurotransmitter in the following:

- Somatic motor fibers to skeletal muscle.
- Preganglionic efferent autonomic fibers of the sympathetic and parasympathetic nervous system.
- Parasympathetic postganglionic fibers.
- Sympathetic postganglionic neurons that innervate sweat glands.

Cholinergic receptors

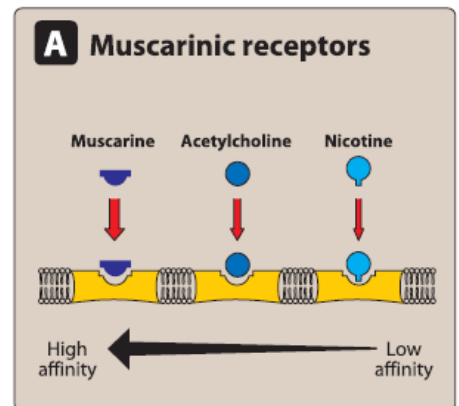
1- Nicotinic receptors

- Subdivided to N1 and N2.
- High affinity to nicotine.
- located in the neuromuscular junction (NMJ) in skeletal muscles, the CNS, the adrenal medulla, and autonomic ganglia.

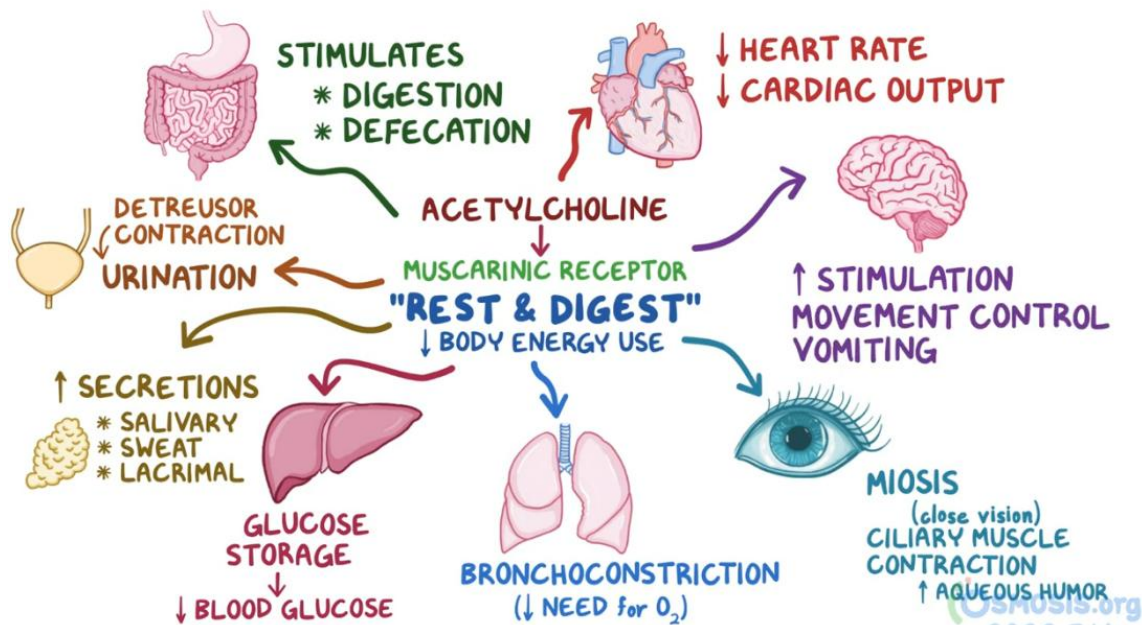


2- Muscarinic receptors.

- Subdivided to M1, M2, M3, M4 and M5.
- High affinity to Muscarine.
- Located on the effector organs.
- M1 in the brain.
- M2 on heart.
- M3 receptors on the bladder, exocrine glands, and smooth muscle.

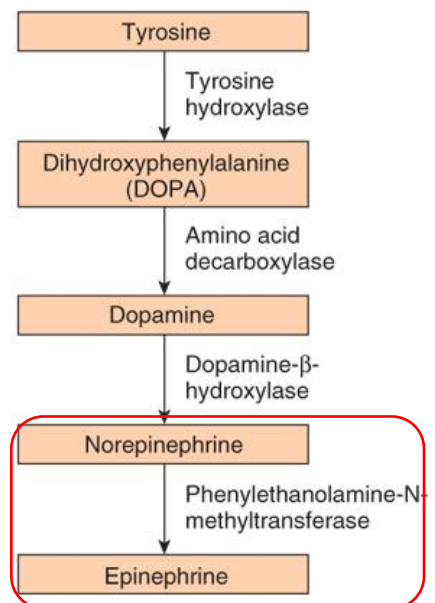


Lippincott Illustrated Reviews: Pharmacology, 7th edition.



Catecholamines

- Catecholamines include norepinephrine (noradrenaline), epinephrine (adrenaline) and dopamine.
- Adrenaline and noradrenaline stimulate adrenergic receptors of SNS.
- Synthesized from a common amino acid **tyrosine**.

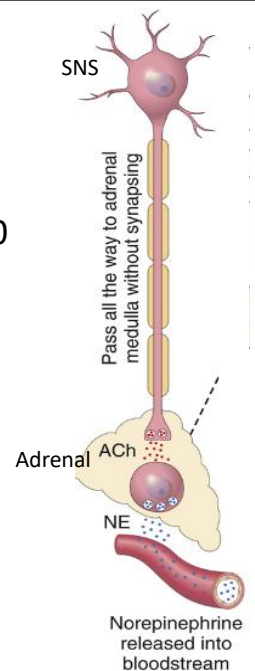


Noradrenaline (Norepinephrine)

- Mediates SNS function of via interacting with adrenergic receptors. Acts as a sympathomimetic or agonist for adrenergic receptors.
- Active in the CNS & PNS. Activates α_1 , α_2 , β_1 receptors.
- Released from most sympathetic postganglionic neurons and by the adrenal medulla (20% of secretion).
- Action is terminated by reuptake into the nerve or diffusion out of the synaptic cleft. It is inactivated via oxidation via monoamine oxidase (MAO) or methylation with catechol-O-methyltransferase (COMT).

Adrenaline (Epinephrine)

- Another catecholamine, secreted by the adrenal medulla (80 % of its secretion).
- Inactivation: oxidation via monoamine oxidase (MAO) or methylation with catechol-O-methyltransferase (COMT).
- Can bind/activate all five alpha and beta receptors.
- It increases the duration of local anesthesia due to its vasoconstrictive effect. However, it can trigger adverse cardiac events via its nonselective adrenergic activity.

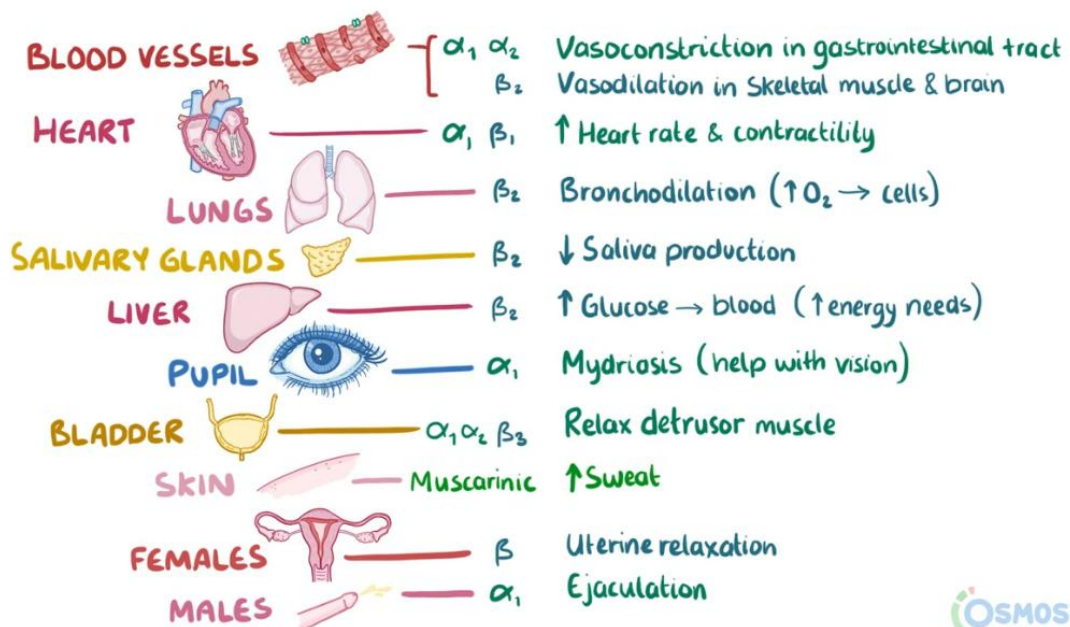


Adrenergic receptors

G-protein coupled receptors

	$\alpha 1$	$\alpha 2$	$\beta 1$	$\beta 2$	$\beta 3$
Main location	Blood vessels (artery and vein), Eye, and bladder	Synaptic nerve ending	Heart	Arteries, bronchi, uterus	Adipose tissue
Function	Vasoconstriction, mydriasis, and increase closure of the internal sphincter of the bladder	prevents norepinephrine and acetylcholine release	increase heart rate and force of contraction.	Vasodilation, bronchodilation. Increase the blood glucose level in blood	increased lipolysis.

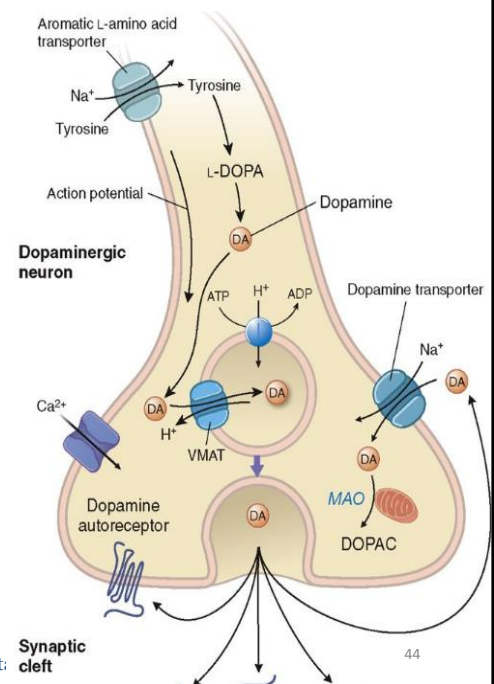
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Dopamine

- Sympathomimetic catecholamine that functions as a CNS neurotransmitter.
- Directly synthesized from L-Dopa and stored.
- Dopamine deficits play a role in the pathogenesis of **depression**.
- Insufficient dopamine biosynthesis in the dopaminergic neurons can cause **Parkinson's disease**.
- Controlled substance like **amphetamine and cocaine** alter the dopamine levels.

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Serotonin

- Serotonin (5-HT) is a monoamine neurotransmitter involved in sleep, eating, arousal, dreaming, mood regulation and temperature/pain transmission.
- In serotonergic neurons, serotonin is synthesized from the amino acid tryptophan, transported to vesicles via VMAT and stored.
- Serotonin is found in high concentrations in enterochromaffin cells throughout the GI tract, in storage granules in platelets (play a role in platelet aggregation & blood clotting), and throughout the CNS.

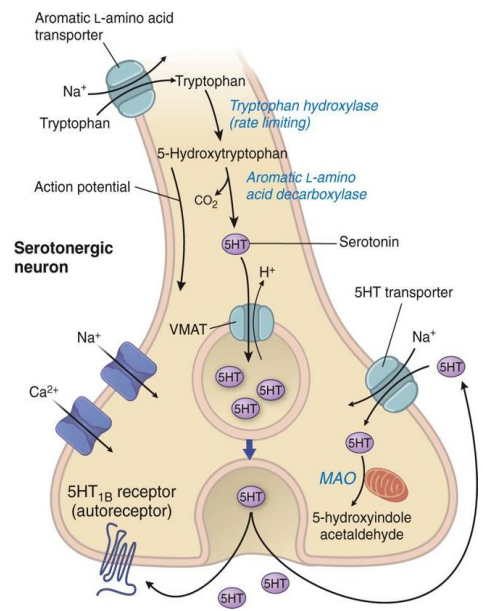
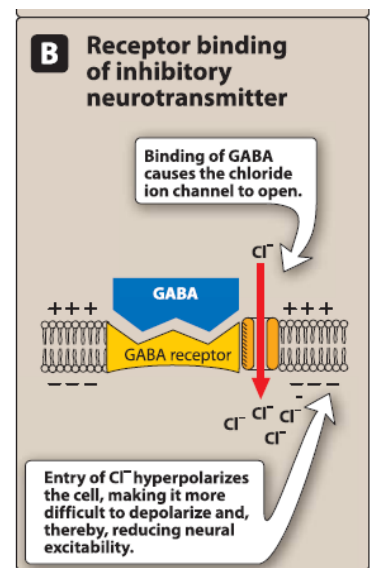


FIGURE 15-2. Presynaptic regulation of serotonin neurotransmission.

γ -aminobutyric acid

- GABA is an inhibitory neurotransmitter in the brain. GABA reduces the excitability of neurons by hyperpolarization of GABA receptors.
- GABA is synthesized from glutamate (aa) via glutamate decarboxylase (GAD; decarboxylation reaction).
- GABA deficiency can be associated with seizures, anxiety.
- Valproic acid, GABA agonist, is used for the treatment of seizures.



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Glycine

- Glycine is another inhibitory neurotransmitter in the CNS.
- Metabolism: Glycine is synthesized from serine by serine hydroxymethyltransferase.
- Receptors: Glycine receptors are mainly distributed in the spinal cord and brain stem.
- Actions of glycine are an active area of research. Strychnine, selective antagonist for glycine, has a potent convulsant effect.

Glutamate

- Glutamate is an excitatory neurotransmitter in the CNS that uses N-methyl-D-aspartate (NMDA) receptors.
- Neurons that release glutamate are called glutamatergic neurons. Glutamatergic neurons are found throughout the brain and spinal cord, and they are termed excitatory neurons.
- Several drugs of abuse modulate glutamic acid and/or GABA to exert tranquilizing or stimulating effects on the brain.
- Glutamate has a leading role in pain transmission from the periphery to the brain.

Around 90 % of the brain neurons use either glutamate or GABA as their neurotransmitter, with the remaining 10 % of neurons using acetylcholine, norepinephrine and other neurotransmitters.

Disorders related to neurotransmitter deficiency

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Seizure and Epilepsy

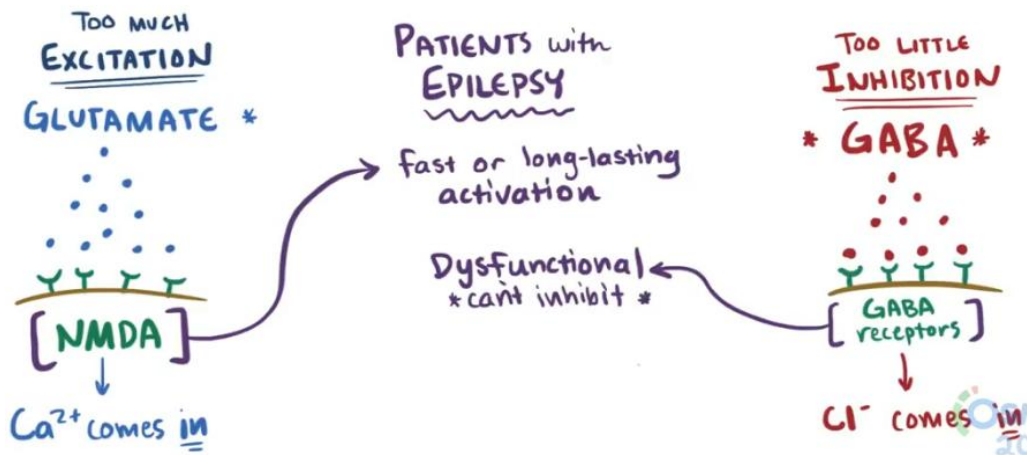
- A seizure is an uncontrolled synchronous firing of neurons leading to abnormal brain electrical activity that may impact the level of consciousness, behavior or memory.
- Epilepsy is a chronic disorder characterized by recurrent unprovoked seizures.
- Around 3 million individuals in the US suffer from epilepsy.
- Over two-thirds of all epileptic seizures begin in childhood

Risk factors:

- **Epilepsy:** history of seizures and genetics can play role.
- **Seizures:** genetic factor can play role. “VITAMINS mnemonic”
 - Vascular (stroke, intracranial hemorrhage),
 - Infection
 - Trauma
 - Autoimmune
 - Metabolic
 - Idiopathic
 - Neoplasm
 - pSychiatric. Family history of seizures (genetic factor) can play role.

Pathophysiology

Abnormal brain electrical activity due increased excitation or decrease inhibition. Clusters of neurons in the brain start sending out lots of excitatory signals, repeatedly.



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Diagnosis:

- **Neurological exam:** Test behavior, movements, mental function ,,,
- **Blood tests:** Detect infections, genetic or metabolic conditions that can cause seizures.
- **Imaging and Electroencephalogram:** detect structural abnormalities and electrical activity in the brain, respectively.

Seizure triggers

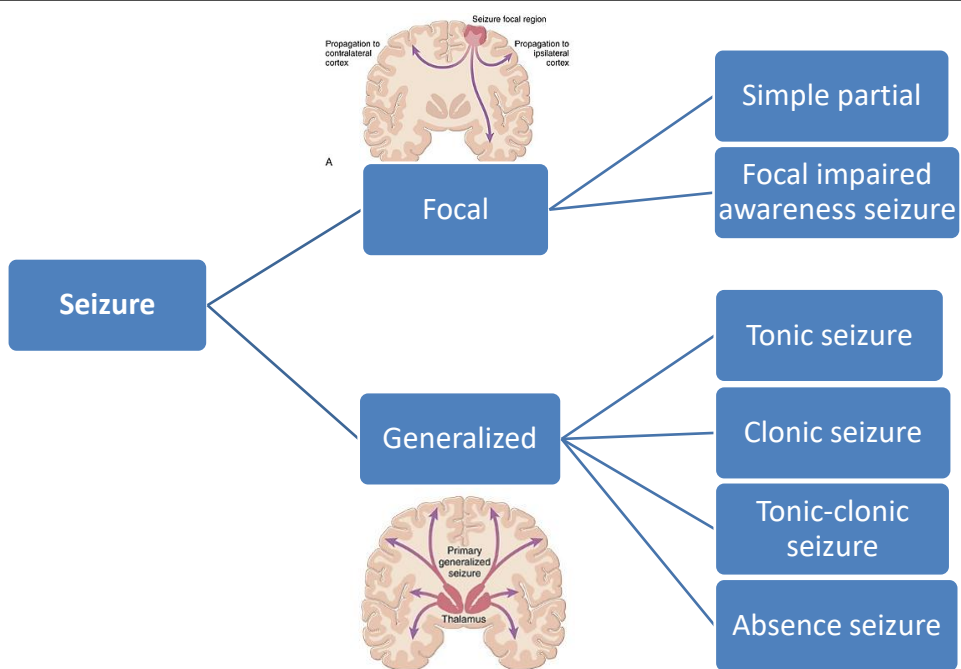
- Lack of sleep. Stress.
- Missed or new medications. Heavy alcohol use or certain illicit drugs (cocaine).
- Flashing lights or loud sounds.
- Hormonal changes (pregnancy,..)

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Symptoms/phases

- Prodromal phase: mostly nonspecific emotional changes such as headache, mood alterations, lethargy. Hours-days before seizure
- Aura: alterations in activity, emotions, hearing, smell, taste or visual perception. Minutes before seizure.
- Ictal: seizure type specific
- Post-ictal: fatigue, confusion, soreness, amnesia, thirst...



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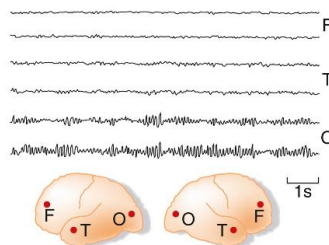
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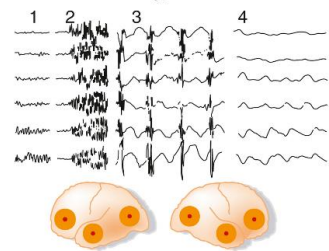
Electroencephalography (EEG) records from frontal (F), temporal (T) and occipital (O) sites on both sides, as shown in the inset diagram.

From Neurological Pathophysiology, second ed. Oxford University Press, New York.

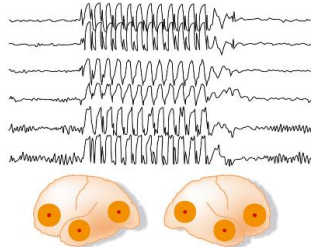
(A) Normal



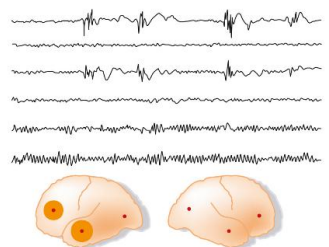
(B) Generalised seizure (grand mal) — tonic-clonic type

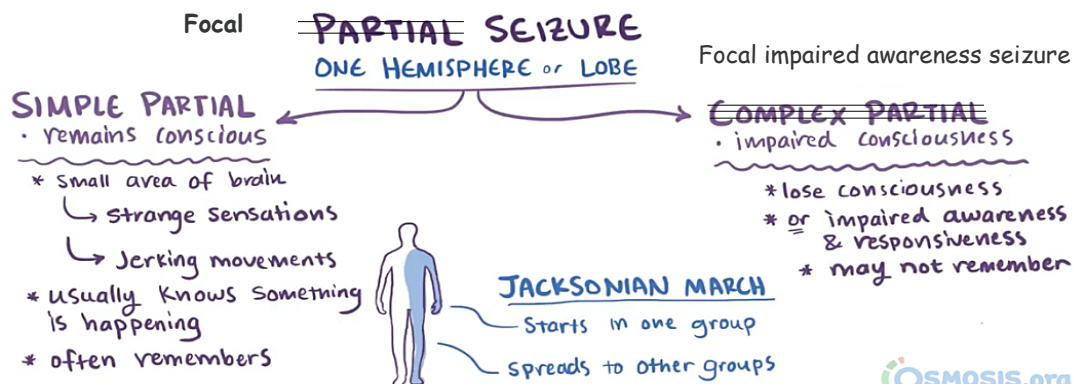


(C) Generalised seizure (petit mal) — absence seizure type



(D) Partial seizure





OSMOSIS.org
2022 Edition

Term	Definition
1- Simple partial seizures	Focal seizures accompanied by <u>preserved consciousness</u> . May be preceded by “an aura” which is a sensory alteration (such as fear, flashing or blurry vision) followed by motor signs.
2- Focal impaired awareness seizure	Focal seizures accompanied by <u>a loss of consciousness</u> . May be preceded by an aura. Duration between 30 second-2 minutes.

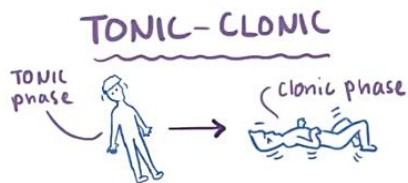
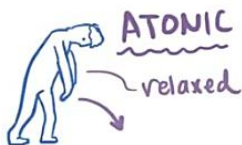
EPILEPSY ~ seizure disorder

↳ recurring & unpredictable
SEIZURES



GENERALIZED SEIZURES

BOTH HEMISPHERES + LOSS OF CONSCIOUSNESS



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Term	Definition
B) Generalized	Electrical discharge that involve the whole brain. Accompanied by loss of consciousness.
1- Tonic	Characterized by continuous muscle contraction of a limb, respiratory, and masticatory muscles. <u>The muscles becomes rigid and flexed</u> which is usually accompanies by the patients <u>falling backward</u> .
2- Atonic	Characterized by <u>muscle relaxation</u> and cause the patients to <u>fall forward</u> .
3- Clonic	<u>Rhythmic violent muscle jerking</u> , without the initial tonic phase.

Term	Definition
B) Generalized	Electrical discharge that involve the whole brain. Accompanied by loss of consciousness.
4- Myoclonic	Characterized by short <u>muscle twitches</u> , either single twitch or several twitches in a short period of time.
5- Tonic-clonic	Initial stiffening and muscle contraction followed by the rythmatic jerking. Most common type of generalized seizures.
6- Absence seizure (aka, petit mal seizures)	<u>Start abruptly. Brief lapses of consciousness</u> (i.e short episode of staring with impaired awareness, spaced-out or daydreaming appearance). Duration 10 seconds or more

STATUS EPILEPTICUS

if SEIZURE lasts ≥ 5 minutes

↳ ongoing or without returning to normal

↳ usually TONIC-CLONIC

Medical emergency

Term	Definition
Status Epilepticus	Medical emergency, characterized by continuous tonic-clonic seizures lasting longer than 5 min. Can lead to airway obstruction, permanent brain injury or death.
Ictal	Events or symptoms that occur during a seizure event.
Postictal	Events or symptoms that occur after the seizure (transition period to normal state, can be associated with confusion, headache, weakness).

First Aid measures

TAKE A.C.T.I.O.N. – Quick Steps!

A-Assess the Situation: Is the person in danger? Remove nearby objects.

C-Cushion the head for protection.

T-Time: Check the time. If the seizure lasts longer than five minutes, call an ambulance.

I-Identity: Look for a medical bracelet or ID card. It may give you information about the person's seizures and particular things to do.

O-Over: Turn the person on his/her side when possible. Stay and provide reassurance.

N-Never restrain the person, put something in his/her mouth, or try to give food or drink.

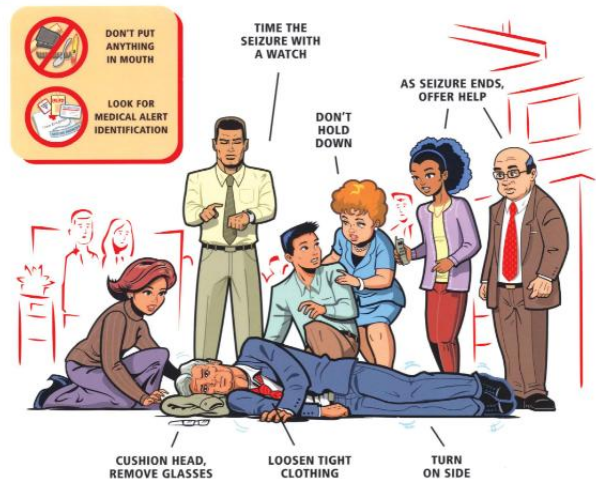
Adapted from

<https://www.epilepsynorcal.org/seizure-first-aid/>

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First Aid for Seizures

(Convulsive, generalized tonic-clonic, grand mal)



An Ambulance Should Be Called If:

- No medical I.D. and no way of knowing whether the seizure is caused by epilepsy.
- Person is pregnant, injured, or diabetic.
- Seizure continues for more than 5 minutes.
- Seizure has happened in water.
- Second seizure starts shortly after the first has ended.
- Consciousness does not return after the shaking has stopped.

Management strategies

1- Anticonvulsants

Drugs reduce seizures through enhancing inhibitory GABAergic activity or modulate glutamate activity (excitatory neurotransmitter).

2- Surgical approach

Focal resection of the epileptogenic cortex (drug-resistant focal epilepsy).

3-Vagus nerve stimulation

It influences neurotransmitter release (precise mechanism is unknown)

4- Ketogenic diet

Forces the body in a state of ketosis which is associated with enhanced GABA levels and reduced neuronal excitability.

Parkinson's disease

- **Classification:** Parkinson's disease (PD) is a progressive neurodegenerative disease with increased risk in the elder population (≥ 65 Y).
 - Two rare forms: Young onset PD (≥ 50 Y) and Juvenile PD (children).
- **Prevalence:** The second most common age-related neurodegenerative disease. Nearly 1 million are living with PD with 90,000 cases diagnosed each year.

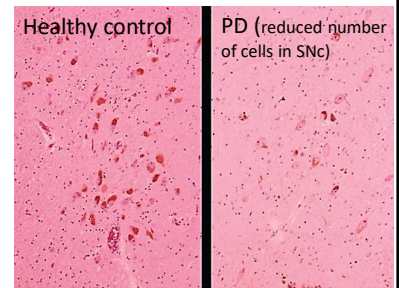
- Risk factors

- Older age. The mean age at diagnosis is 70.5 years. Aging is the primary risk factor.
- Family history of PD. Genetic, certain gene mutations can increase the risk for PD, like mutations in *parkin* gene.
- Male gender. PD incidence is higher in men compared to women at all ages.
- Environmental factors like pesticide exposure, head trauma or injury that damage dopaminergic neurons.

Pathophysiology:

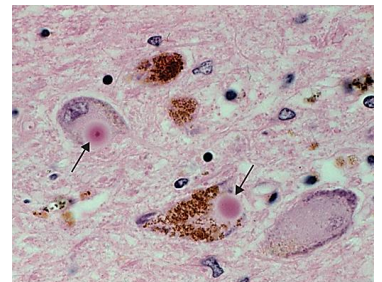
PD is attributed to the degeneration of dopaminergic neurons in the substantia nigra pars compacta (SNc). Causes depletion of dopamine from the substantia nigra & disruption of the thalamus-motor cortex connection.. Clinical symptoms are clear after >60% of dopaminergic neurons are depleted.

Microscopically, Lewy bodies (arrows) are present in the affected substantia nigra neurons. Lewy bodies are abnormal protein aggregates. Decreased dopamine levels can cause the upregulation of acetylcholine levels in PD patients.



B

Source: Joseph Loscalzo, Anthony Fauci, Dennis Kasper, Stephen Hauser, Dan Longo, J. Larry Jameson: Harrison's Principles of Internal Medicine, 21e Copyright © McGraw Hill. All rights reserved.



C

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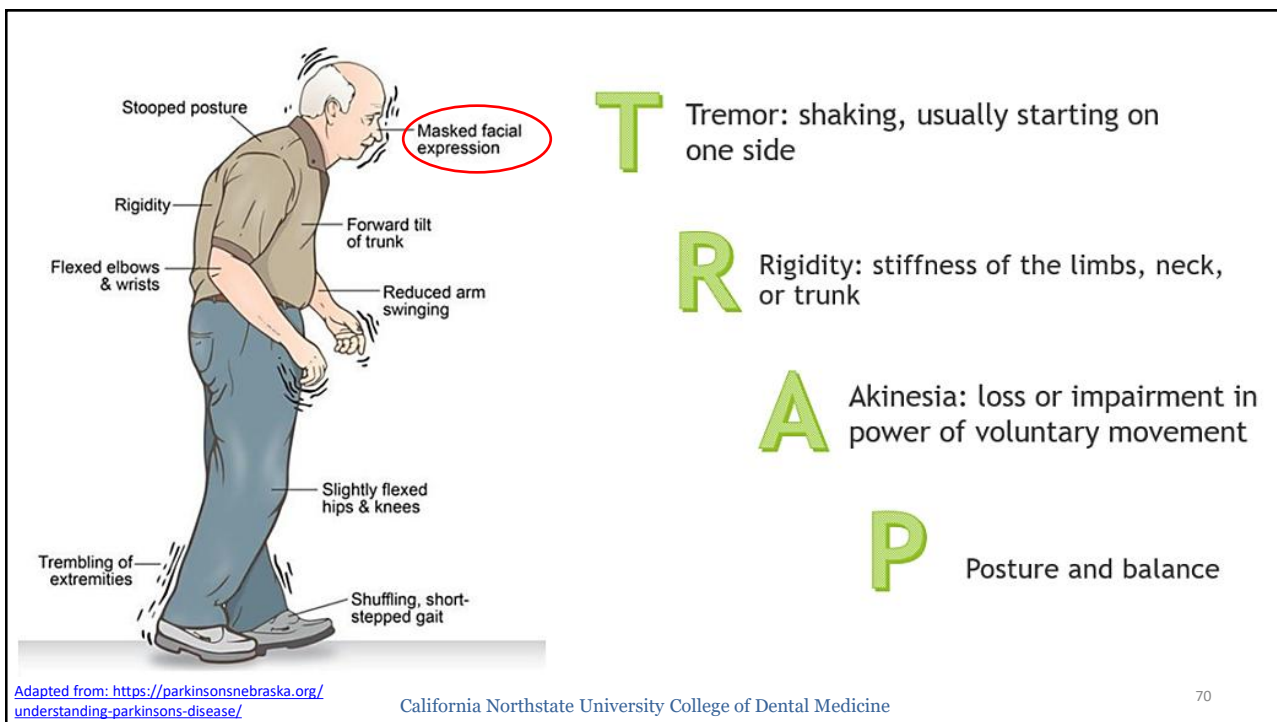
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Clinical manifestation

- 1- **Tremors at rest:** Tremors are most noticeable when the body part is supported by gravity.
- 2- **Rigidity:** increased muscle stiffness. Occur in 75-90 % of patients. One of the early symptoms (cogwheeling or lead pipe rigidity).
- 3- **Bradykinesia:** generalized slowness of movement, present at onset of PD in around 80 % of patients and worsens with the disease progression.
- 4- **Postural instability:** impairment of postural reflexes that increases the tendency to fall with a significant risk of injury.

- 5- Other motor defects can include akinesia, masked facial expression (hypomimia), speech difficulty.
- 6- Non motor defects include: Cognitive dysfunction, dementia and mood disorders.

Prognosis: As the disease progresses, patients might develop difficulty in swallowing. Trouble swallowing can cause the leakage of food/liquid to the lungs causing aspiration pneumonia. Aspiration pneumonia is the leading cause of death in Parkinson's.



Management

Chronic, progressive with no cure

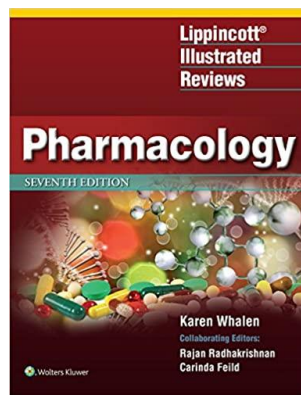
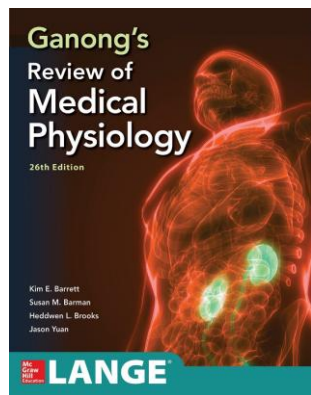
1- Nonpharmacological

Patient education, regular exercise, social and emotional support and dietary changes to prevent malnutrition.

2- Pharmacological

The therapeutic goal aims to increase the dopamine availability (via dopamine precursors or prevent dopamine degradation) and/or control tremors (via antimuscarinic medications).

Recommended resources



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